

# RAKESH MANDAL

PhD Scholar, Department of Applied Mechanics, Indian Institute of Technology Delhi

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## RESEARCH SUMMARY

I develop generative AI models for engineering design that produce high-performing structures in seconds and stay reliable under manufacturing uncertainty. My PhD work uses flow matching, neural surrogates and robust optimisation to replace slow, solver-in-the-loop design with learned generative priors, for 2D and 3D topology optimisation, automotive wheel design and aerodynamic shape design. Before my PhD, I worked as a production engineer at Hero MotoCorp, where I saw first-hand how manufacturing variation turns good designs into rejected parts.

**Interests:** generative models (flow matching, diffusion) · topology and shape optimisation · robust and reliability-based design under uncertainty · neural surrogates and operator learning · energy-efficient and multimodal generative design

## EDUCATION

**PhD (Direct, after B.Tech.), Applied Mechanics**

Jul 2025 – present (expected 2029)

Indian Institute of Technology (IIT) Delhi

New Delhi

- ▶ Advisor: Prof. Souvik Chakraborty
- ▶ Thesis (working title): *Multimodal, Robust, Sustainable and Reliability-Aware Generative Design*

**B.Tech., Mechanical Engineering**

2018 – 2022

National Institute of Technology (NIT) Durgapur

Durgapur, West Bengal

- ▶ CGPA 8.24/10 (final-semester SGPA 9.44/10)
- ▶ Final-year project: *Observing and Modelling Nonlinear Dynamics in an Internal Combustion Engine*, under Prof. Sirshendu Mondal

Class XII (WBCHSE), 2018: 83.8%   ·   Class X (WBBSE), 2016: 84.57%, first position in school

## PUBLICATIONS

\*Equal contribution.

### Journal articles under review

- [J1]   **R. Mandal\***, A. P. Padhi\*, R. Nayek, A. Dasgupta, S. Chakraborty. *A Conditional Flow-Matching Framework for Generative Topology Optimization in Two and Three Dimensions*. Under review, *Computer Methods in Applied Mechanics and Engineering* (CMAME).

### Manuscripts in preparation

- [M1]   **R. Mandal**, S. M. Pekhale, S. Chakraborty. *RobustFlow: A Unified Flow-Matching Architecture for Robust Design Optimization, from 2D Topology Optimization to 3D Automotive Wheel Design*. In preparation for *Structural and Multidisciplinary Optimization*.
- [M2]   S. Singh, **R. Mandal**, S. Chakraborty. *Flow Matching for Design Under Uncertainty: A Generative Framework for Reliable Shape Design*. In preparation for *Computer Physics Communications*.

## RESEARCH EXPERIENCE

**Doctoral Researcher**

Jul 2025 – present

Department of Applied Mechanics, IIT Delhi (advisor: Prof. Souvik Chakraborty)

New Delhi

**TopoFlow: generative topology optimisation in 2D and 3D** [J1]

- ▶ Built a single conditional flow-matching model (U-Net velocity field with Fourier-feature and FiLM conditioning) that generates 2D and 3D optimised structures without architectural changes.
- ▶ 2D: matches classical SIMP compliance (median ratio 1.008), with 93.2% of designs within 10% of the SIMP reference, at 25–100× lower cost.
- ▶ 3D (SELTO benchmarks): up to 96.5% of designs within 10% of the FEM reference compliance.

**RobustFlow: robust design optimisation with flow matching** [M1]

- ▶ Designed a two-stage architecture: a flow-matching prior, then robust fine-tuning against a pluggable physics critic (a differentiable SIMP solver or a learned surrogate).
- ▶ 2D topology optimisation: reduced mean compliance by up to 67% and its standard deviation by up to 18.8× versus an unregularised generator.
- ▶ 3D automotive wheels (904 industrial CAD models with FE and modal data): surrogate  $R^2 \geq 0.89$ ; generated wheels reach target eigenfrequencies with population coefficient of variation below 0.4%.

**FM-DUF: design under fabrication uncertainty** [M2]

- Hierarchical flow-matching model of nominal designs and their fabrication-induced geometric variations, coupled with a wavelet neural operator surrogate and Bayesian optimisation for quantile-based robust design of airfoils and optical metasurface absorbers.

## INDUSTRY EXPERIENCE

### Graduate Engineer Trainee (Assistant Manager role), Production

Jul 2022 – 2025

Hero MotoCorp Ltd.

Halol Plant, Gujarat

- Led the team that reduced crankshaft rejections in the induction-hardening operation by 70%.
- Planned daily production and manpower; ran cost-optimisation and rejection-reduction projects using structured problem solving on the shop floor.
- Pillar head of the GreenCo (green company) initiative; deputy leader of the Development Management pillar of Total Productive Maintenance (TPM); strengthened the 5S red-tag system.

## UNDERGRADUATE PROJECTS AND TRAINING

- **Nonlinear dynamics in an IC engine** (B.Tech. project, Prof. Sirshendu Mondal): observed and modelled nonlinear combustion dynamics.
- **Cellular-automaton model of particle motion and its applications** (mini project, Prof. Sirshendu Mondal).
- **SAE BAJA, Team NDORS, NIT Durgapur**: team member at SAE BAJA 2020 (Pithampur, Indore); Technical Head, Powertrain, 2021.
- **Industrial internship, MSME-CTTC Bhubaneswar** (2 weeks): turning, milling, drilling and CNC machining; built a hand-injection mould; CATIA training.

## SKILLS

|                                |                                                                                                                                                                                        |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Machine learning</b>        | PyTorch (mixed-precision GPU training), flow matching, diffusion models, 2D/3D U-Nets, FiLM and classifier-free guidance, neural operators, surrogate modelling, Bayesian optimisation |
| <b>Computational design</b>    | Topology optimisation (SIMP), finite-element analysis, robust design optimisation, uncertainty quantification, SDF/voxel geometry processing                                           |
| <b>Programming &amp; tools</b> | Python (NumPy, SciPy, pandas, trimesh, scikit-image), MATLAB, Git, $\LaTeX$ , Linux GPU workstations (NVIDIA A40), Weights & Biases                                                    |
| <b>CAD</b>                     | CATIA, SolidWorks, AutoCAD                                                                                                                                                             |
| <b>Languages</b>               | English, Hindi, Bengali                                                                                                                                                                |

## LEADERSHIP AND ACHIEVEMENTS

- Led the induction-hardening rejection-reduction project at Hero MotoCorp (70% reduction).
- Technical Head, Powertrain, Team NDORS (SAE BAJA), NIT Durgapur, 2021.
- Admitted directly to the PhD programme at IIT Delhi after the B.Tech.
- First position in school, West Bengal Secondary (Madhyamik) Examination, 2016.